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Locking structure, housing assembly, and electronic device

Abstract

Provided are a locking structure, a housing assembly, and an electronic device. The locking structure includes first and second structural bodies, a pressing member, and a clamped member. The first structural body is provided with first and second mounting portions. The second structural body includes a locking portion. The pressing member is movably provided at the first mounting portion. The clamped member is movably provided at the second mounting portion, and can be engaged with the locking portion. In mounting the second structural body to the first structural body, the clamped member is driven by the locking portion to move, and is restored so as to be engaged with the locking portion. In detaching the second structural body from the first structural body, the pressing member is pressed by an external force to drive the clamped member to move so as to be disengaged from the locking portion.

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Background/Summary

CROSS-REFERENCE OF RELATED APPLICATIONS (1) This application is a continuation of International Application No. PCT/CN2021/136621, filed Dec. 9, 2021, which claims priority to: Chinese Patent Application No. 202120141448.0, filed Jan. 19, 2021; Chinese Patent Application No. 202110069698.2, filed Jan. 19, 2021; and Chinese Patent Application No. 202120140667.7, filed Jan. 19, 2021. The entire disclosures of the aforementioned applications are incorporated herein by reference.

TECHNICAL FIELD
(1) The present disclosure relates to the field of wearable devices, and in particular to a locking structure, a housing assembly and an electronic device.
BACKGROUND
(2) Smart wearable devices are capable of being worn on the body, and they may detect a health condition at any time, perform instant messaging, and offer other functions. In recent years, more and more people have begun to pay attention to and use the smart wearable devices. A traditional smart watch includes a watch body and a watch strap connected to the watch body, and the watch strap may be wound around and fixed on the wrist of the user.
(3) The traditional watch strap is connected to the watch body by means of mating between a slot and a spring bar. The watch strap may be connected stably with the watch body, after two ends of the spring bar are inserted into the slot. However, due to the small and special structure of the spring bar, if one desires to detach the spring bar from the slot or install the spring bar into the slot, it is necessary to use a special tool such as a spring-bar needle for the detachment and installation, which is inconvenient to operate.
SUMMARY
(4) Embodiments of the present disclosure provide a locking structure, a housing assembly and an electronic device.
(5) In a first aspect, an embodiment of the disclosure provides a locking structure. The locking structure includes a first structural body, a second structural body, a pressing member and a clamped member. The first structural body is provide with a first mounting portion and a second mounting portion, and the second structural body includes a main body and a locking portion protruding relative to the main body. The pressing member is movably provided at the first mounting portion, and is configured to move relative to the first mounting portion under an external force. The clamped member is movably provided at the second mounting portion and is positioned at the first structural body through the second mounting portion, and the clamped member is configured to be engaged with the locking portion. At least partial structure of the clamped member is arranged opposite to the pressing member. In a process of mounting the second structural body to the first structural body, the clamped member is driven by the locking portion to move relative to the second mounting portion, and is restored so as to be engaged with the locking

portion. In a process of detaching the second structural body from the first structural body, the pressing member is capable of being pressed to drive the clamped member to move so as to enable the clamped member to be disengaged from the locking portion.
(6) In a second aspect, an embodiment of the present disclosure further provides a housing assembly. The housing assembly includes a middle frame, a housing, and a key. The housing is provided with a first mounting portion, and the middle frame is provided with a second mounting portion. The key is connected to the middle frame and is configured to connect the middle frame and an external structure including a locking portion. The key includes a pressing member and a clamped member. The pressing member is movably provided at the first mounting portion, and is configured to move relative to the first mounting portion in a first direction under an external force. The clamped member is movably provided at the second mounting portion and is positioned at the middle frame through the second mounting portion, and the clamped member is configured to be engaged with the locking portion of the external structure. At least partial structure of the clamped member is arranged opposite to the pressing member. In a process of mounting the external structure to the middle frame, the external structure is adapted to move in a second direction to push the clamped member to move relative to the middle frame in the first direction, and the clamped member is capable of being restored so as to be engaged with the locking portion of the external structure. In a process of disengaging the external structure from the middle frame, the pressing member is capable of being pressed to drive the clamped member to move in the first direction so as to be disengaged from the external structure, where the first direction is different from the second direction.
(7) In a third aspect, an embodiment of the present disclosure provides an electronic device. The electronic device includes a casing, a key, and a clamped member. The casing is provided with a first mounting portion and a second mounting portion communicated with each other. The key includes a pressing member and an elastic connection member, the elastic connection member is connected to the pressing member, and the pressing member is provided at the first mounting portion. The clamped member is provided at the second mounting portion, and the assembling and positioning of the clamped member to the casing are performed separately from the assembling and positioning of the pressing member to the casing. The clamped member is adapted to mate with a locking portion of the strap, to make the strap confined at the casing. When the strap is confined at the casing, the elastic connection member abuts against the casing, to make the pressing member confined at the casing. The pressing member is adapted to, when being pressed, move relative to the casing, and drive the clamped member to move, to enable the clamped member to be disengaged from the strap, and enable the strap to be detached from the casing.
(8) Other features and aspects of the disclosed features will become apparent from the following detailed description, taken in conjunction with the accompanying drawings, which illustrate, by way of example, the features in accordance with embodiments of the disclosure. The summary is not intended to limit the scope of any embodiments described herein.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) In order to more clearly explain technical solutions of embodiments of the present disclosure, drawings used in the embodiments will be briefly introduced below. Obviously, the drawings as described below are merely some embodiments of the present disclosure. Based on these drawings, other drawings can be obtained by those skilled in the art without inventive effort.
(2) FIG. **1** is a partial perspective view illustrating a locking structure according to an embodiment of the present disclosure.
(3) FIG. **2** is an exploded view illustrating the locking structure illustrated in FIG. **1** from a first

perspective.
(4) FIG. **3** is a perspective view illustrating an assembly of a housing and a pressing member of the locking structure illustrated in FIG. **1**.
(5) FIG. **4** is a perspective view illustrating the assembly of the housing and the pressing member illustrated in FIG. **3** from another perspective.
(6) FIG. **5** is an exploded view illustrating the housing and the pressing member illustrated in FIG. **3**.
(7) FIG. **6** is an orthogonal view illustrating an assembly of a first structural body and a key, which mate with each other, of the locking structure illustrated in FIG. **1**.
(8) FIG. **7** is a perspective view illustrating the key of the locking structure illustrated in FIG. **2**.
(9) FIG. **8** is a schematic cross-sectional view illustrating mating of the key illustrated FIG. **7** with the housing.
(10) FIG. **9** is an exploded view illustrating the first structural body of the locking structure illustrated in FIG. **1**.
(11) FIG. **10** is a cross-sectional view illustrating the locking structure illustrated in FIG. **1**.
(12) FIG. **11** is an exploded view illustrating a connecting assembly and the first structural body illustrated in FIG. **8**.
(13) FIG. **12** is an exploded view illustrating the connecting assembly and the first structural body illustrated in FIG. **11**, from another perspective.
(14) FIG. **13** is a partial cross-sectional view illustrating an elastic connection member of the locking structure illustrated in FIG. **11**.
(15) FIG. **14** is an exploded view illustrating the locking structure illustrated in FIG. **1** from a second perspective.
(16) FIG. **15** is an exploded view illustrating the locking structure illustrated in FIG. **1** from a third perspective.
(17) FIG. **16** is another cross-sectional view illustrating the locking structure illustrated in FIG. **1**.
(18) FIG. **17** is a schematic perspective view illustrating an electronic device according to an embodiment of the present disclosure.
(19) FIG. **18** is a schematic perspective view illustrating a smart wearable device according to an embodiment of the present disclosure.

DETAILED DESCRIPTION OF PREFERRED EMBODIMENTS

(20) The technical solutions in the embodiments of the present disclosure will be clearly and comprehensively described below with reference to the accompanying drawings in the embodiments of the present disclosure. Apparently, the described embodiments are only some of the embodiments of the present disclosure, but not all of the embodiments. All other embodiments, obtained by those skilled in the art without inventive work based on the embodiments in the present disclosure, fall within the protection scope of the present disclosure.
(21) A traditional smart watch includes a watch body and a watch strap connected to the watch body, and the watch strap may be wound around and fixed on the wrist of the user. The traditional watch strap is connected to the watch body by means of mating between a slot and a spring bar. The watch strap may be connected stably with the watch body, after two ends of the spring bar are inserted into the slot. However, due to the small and special structure of the spring bar, if one desires to detach the spring bar from the slot or install the spring bar into the slot, it is necessary to use a special tool such as a spring-bar needle for the detachment and installation, which is inconvenient to operate.
(22) In order to solve the above problem, the embodiments of the present disclosure provide a locking structure, a smart wearable device, and a housing assembly. The locking structure includes a first structural body, a second structural body, a pressing member and a clamped member. The first structural body is provided with a first mounting portion and a second mounting portion. The second structural body includes a main body and a locking portion protruding relative to the main

body. The pressing member is movably provided at the first mounting portion, and configured to move relative to the first mounting portion under an external force. The clamped member is movably provided at the second mounting portion, and is positioned at the first structural body through the second mounting portion. The clamped member is configured to be engaged with a locking portion. At least partial structure of the clamped member is arranged opposite to the pressing member. In a process of mounting the second structural body to the first structural body, the clamped member is driven by the locking portion to move relative to the second mounting portion, and is restored so as to be engaged with the locking portion. In a process of detaching the second structural body from the first structural body, the pressing member is capable of being pressed to drive the clamped member to move, so as to enable the clamped member to be disengaged from the locking portion.

(23) In the locking structure provided by the embodiments of the present disclosure, the pressing member and the clamped member are configured to be connected between the first structural body and the second structural body, and the clamped member can be configured to be engaged with the locking portion of the second structural body. With the structures of the clamped member and the locking portion which are movably engaged, the second structural body is enabled to be conveniently connected to and detached from the first structural body. For example, in the process of mounting the second structural body to the first structural body, the clamped member is driven by the locking portion to move relative to the first structural body, and is restored so as to be engaged with the locking portion; in this way, the second structural body is enabled to be conveniently mounted to the first structural body. For another example, in the process of detaching the second structural body from the first structural body, the pressing member can be pressed to drive the clamped member to move so as to be disengaged from the locking portion; in this way, the second structural body is enabled to be conveniently detached from the first structural body. When the locking structure is applied to a smart wearable device, a detachable connection of the device body with a wearable part can be conveniently achieved by means of the locking structure.

(24) The technical solutions in the embodiments of the present disclosure will be clearly and comprehensively described in connection with the accompanying drawings in the embodiments of the present disclosure.

(25) As illustrated in FIG. **1**, an embodiment of the present disclosure provides a locking structure **100** for connecting two parts, which enables the two parts to be detachably connected together, and enables the two parts connected to be easily detached from each other. The locking structure **100** includes a first structural body **10**, a second structural body **30**, and a connecting assembly **50**. The connecting assembly **50** is connected between the first structural body **10** and the second structural body **30**, for realizing the detachable connection of the first structural body **10** and the second structural body **30**. In a specific application embodiment, the first structural body **10** may be provided on one of two parts that need to be connected, and the second structural body **30** may be provided on the other one of the two parts that need to be connected; as such, the two parts may be detachably connected. The first structural body **10** and the second structural body **30** may be structural components of the parts that need to be connected, or may be attached to structures of the parts that need to be connected, which is not limited in the present disclosure, and thus the locking structure **100** can be widely applied.

(26) As illustrated in FIG. **2**, in the embodiment of the present disclosure, the first structural body **10** is provided with a first mounting portion **101** and a second mounting portion **103**. The first mounting portion **101** and the second mounting portion **103** are configured to mount the connecting assembly **50**. The specific structures of the first mounting portion **101** and the second mounting portion **103** are not limited. For example, the first mounting portion **101** may include one or more of a mounting groove, a mounting hole, a snap-fit structure or other mounting and receiving structures, to facilitate the mounting of the connecting assembly **50**. Similarly, the second mounting portion **103** may include one or more of a mounting groove, a mounting hole, a snap-fit structure or

other mounting and receiving structures, to facilitate the mounting of the connecting assembly **50**. The structure of the second mounting portion **103** may be the same as or different from the structure of the first mounting portion **101**. For example, the first mounting portion **101** includes a key groove **1011**, and the second mounting portion **103** may include a groove **1031** communicated with the key groove; where the key groove **1011** extends to an outer surface of a housing **14**, that is, the key groove **1011** extends to a surface of the housing **14** for the user to touch. The groove **1031** may be provided in a middle frame **12**, and the second structural body **30** may be mounted to an electronic device **500** (as illustrated in FIG. **17**) through the groove **1031**, and can be detached from the groove **1031**. Specifically, in the embodiment, the groove **1031** may be used as a reference, two grooves **1031** are oppositely provided at two ends of the electronic device **500** in a length direction thereof, and a display screen module **120** and the housing **14** of the electronic device **500** are respectively located at two ends of the electronic device **500** in a thickness direction thereof. For simplicity, the length direction of the electronic device **500** may be regarded as a second direction, and the thickness direction of the electronic device **500** may be a first direction, that is, the first direction and the second direction are perpendicular to each other. The second structural body **30** can be installed into the groove **1031** along the second direction, and is confined at the first structural body **10**. In the embodiments of the present disclosure, the first direction and the second direction are only used as a reference, for clearly and concisely describing the positions of the second structural body **30** and other parts relative to the middle frame **12**, and this should not be regarded as strictly limiting the technical solution. For example, in other embodiments, there is no need to take the length direction of the electronic device **500** as the second direction or take the thickness direction of the electronic device **500** as the first direction; and in other embodiments, the first direction and the second direction may be at an acute angle.

(27) The second structural body **30** includes a main body **32** and a locking portion **34** protruding from the main body **32**. The locking portion **34** is configured to be engaged with the connecting assembly **50**, for the detachable connection between the second structural body **30** and the first structural body **10**.

(28) The connecting assembly **50** includes a key **54**. The key **54** is configured to connect the first structural body **10** and the second structural body **30**. The key **54** includes a pressing member **541** and a clamped member **543**. The pressing member **541** is movably provided at the first mounting portion **101**, and is configured to move relative to the first mounting portion **101** (for example, move in the first direction X) under an external force. In the embodiment of the present disclosure, the pressing member **541** is positioned at the first structural body **10** through the first mounting portion **101**. The clamped member **543** is positioned at the first structural body **10** through the second mounting portion **103**, and is configured to be engaged with the locking portion **34**. At least part of the structure of the clamped member **543** is arranged opposite to the pressing member **541**. In the process of mounting the second structural body **30** to the first structural body **10**, the clamped member **543** is driven by the locking portion **34** to move relative to the second mounting portion **103**, and is restored so as to be engaged with the locking portion **34**. In a process of detaching the second structural body **30** from the first structural body **10**, the pressing member **541** can be pressed to drive the clamped member **543** to move so as to be disengaged from the locking portion **34**. In the locking structure **100**, the key **54** is connected between the first structural body **10** and the second structural body **30**. While the key **54** is provided at the first structural body **10**, the clamped member **543** thereof can be engaged with the locking portion **34** of the second structural body **30**. The pressing member **541** may be utilized to drive the clamped member **543** to be movably engaged with the locking portion **34**. Thus, the second structural body **30** can be conveniently connected to and detached from the first structural body **10**.

(29) In the embodiment of the present disclosure, with regard to the structure that the pressing member **541** and the clamped member **543** are arranged opposite to each other, it is not limited to one that the pressing member and the clamped member are directly opposite to and spaced apart

from each other. In some embodiments, the pressing member **541** and the clamped member **543** may be opposite to each other in such a manner that, for example, at least partial structure of the pressing member **541** and at least partial structure of clamped member **543** are spaced apart from each other, or a middle element or component may be provided between the pressing member **541** and the clamped member **543**. In some other embodiments, the pressing member **541** and the clamped member **543** may be opposite to each other and overlapped with or attached to each other, for example, at least partial structure of the pressing member **541** may abut against or be in surface contact with at least partial structure of the clamped member **543**. In some further embodiments, there is a preset gap between the pressing member **541** and at least partial structure of the clamped member **543**, and the width of the gap should be in a preset range. For example, the width of the gap may be greater than or equal to 0.1 mm and less than or equal to 1.5 mm. When the pressing member **541** is pressed by an external force, the pressing member **541** can push the clamped member **543** to move only if a distance that the pressing member **541** moves is greater than the width, thereby preventing disengagement of the clamped member **543** from the locking portion **34** that is caused when the pressing member **541** is pressed by mistake. As such, by making the pressing member **541** and at least partial structure of the clamped member **543** arranged opposite to each other, the pressing member **541** can be pressed to drive the clamped member **543** to move, that is, transmission of movement is enabled between the pressing member and the clamped member by means of the opposite position relationship thereof.

(30) In the case where the pressing member **541** and the clamped member **543** are arranged opposite to each other, in order to ensure that the pressing member **541** can reliably drive the clamped member **543** to move, the pressing member **541** is positioned at the first mounting portion **101** or positioned at the first structural body **10** through the first mounting portion **101**, and the clamped member **543** is positioned at the second mounting portion **103** or positioned at the first structural body **10** through the second mounting portion **103**. In this specification, with regard to expression that a component is “positioned at” another component, it may be understood that the position of the former relative to the latter is generally defined through the structure of the latter itself or through a middle element, such that the former is movable within a predetermined range relative to the latter. For example, regarding the pressing member **541** being positioned at the first mounting portion **101**, it may be understood that the pressing member **541** is limited within a predetermined movement range by means of the structure of the first mounting portion **101** or/and the third party component; alternatively, regarding the pressing member **541** being positioned at the first structural body **10** through the first mounting portion **101**, it is understandable that the pressing member **541** is limited within the predetermined movement range by means of the structure of the first mounting portion **101**, or it may be understood that the pressing member **541** is limited within the predetermined movement range by means of mating of the structure of the first mounting portion **101** with the structure(s) of the pressing member **541** or/and a third party component. The first mounting portion **101** for limiting the pressing member **541** may include a position-limiting structure such as a hole structure, a groove structure, and a guide rail structure. Through position-limiting cooperation of such position-limiting structure with the structure of the pressing member **541**, the pressing member **541** is positioned. For another example, regarding the clamped member **543** being positioned at the first structural body **10** through the second mounting portion **103**, it is understandable that the clamped member **543** is limited within a predetermined movement range by means of the structure of the second mounting portion **103**, where the second mounting portion **103** for limiting the clamped member **543** may include a position-limiting structure such as a hole structure, a groove structure, and a guide rail structure. Through position-limiting cooperation of such position-limiting structure with the structure of the clamped member **543**, the clamped member **543** is positioned. The position-limiting structure and positioning structure will be described in detail hereafter in the specification, according to some particular embodiments of the present disclosure.

(31) In this way, at the time of assembling the pressing member **541** and the clamped member **543**, the pressing member **541** can be positioned and mounted through the first mounting portion **101**, and the clamped member **543** can be positioned and mounted through the second mounting portion **103**, which avoids a fitting issue in installation that is caused due to a too long dimension chain generated when the pressing member **541** and the clamped member **543** are fixedly connected through other fastening structures. For example, in the related art, at the time of installing a key, multiple parts (such as a pressing portion and an abutting portion) of the key itself are firstly assembled together, and then the assembled key is mounted onto the first structural body. In this case, when any one of the pressing member and the clamped member has a manufacturing tolerance, the assembly obtained by firstly assembling the two parts together has a long dimension chain, which would inevitably cause a large installation error. When the whole of the two parts needs to be mounted onto the first structural body, the overall large installation error causes the overall dimension of the two parts to be extremely likely not to meet the assembly tolerance requirement of the whole and the first structural body, and the probability of failing to performing the mounting is greatly increased because the whole interferes with other parts of the first structural body. Therefore, there are high accuracy requirements on the manufacturing dimensions of the parts (such as the pressing member and the clamped member), and it is very produce and assemble. However, in this disclosure, the key **54** is designed as a pressing member **541** and a clamped member **543** which are substantially separated, and the pressing member **541** and the clamped member **543** can be respectively provided at the first mounting portion **101** and the second mounting portion **103** of the first structural body **10**, so that the assembly requirements of positioning and assembling an individual part need to be met. There is no longer a dimension chain for assembly between the pressing member **541** and the clamped member **543**, and there is a short dimension chain generated when the key **54** is mounted onto the first structural body **10**, which reduces the difficulty of producing and assembling the locking structure **100**.
(32) The locking structure **100** provided in the present disclosure will be described in detail below with reference to the embodiments illustrated in the drawings.
(33) In the embodiment illustrated in FIG. **2**, the first structural body **10** includes a middle frame **12** and a housing **14** connected to the middle frame **12**. In the embodiment, the housing **14** is substantially covered on the middle frame **12**, and is configured to define a main appearance surface of the first structural body **10**. The first mounting portion **101** is provided at the housing **14**, and the second mounting portion **103** is provided at the middle frame **12**. The middle frame **12** is configured to define a main supporting structure. For example, the middle frame **12** may be configured to define a supporting structure for pats connected thereto, so as to receive electronic components of the parts. For ease of description, in the specification, a first direction X, a second direction Y, and a third direction Z are defined with reference to a length direction, a width direction, and a height direction of the middle frame **12**. As illustrated in FIG. **2**, the first direction X is defined as the thickness direction of the middle frame **12**, and the positive direction of the first direction X faces downward in the figure; the second direction Y is defined as the length direction of the middle frame **12**, and the positive direction of the second direction Y faces towards the upper left in the figure; and the third direction Z is defined as the width direction of the middle frame **12**, and the positive direction of the third direction Z faces towards the lower left in the figure. Further, the first direction X, the second direction Y, and the third direction Z may be perpendicular to each other. The middle frame **12** may be provided on a part to which the middle frame **12** needs to be connected, or may be an integral part of this part. For example, when the locking structure **100** is applied to a smart watch, the middle frame **12** may be provided at the body of the smart watch, or may be a part of the body. The shape of the middle frame **12** is not limited, and it may be a rectangular frame, a circular frame, or other block objects.
(34) As illustrated in FIG. **3**, in the embodiment, the first mounting portion **101** may include an accommodating groove **141** provided in the housing **14**, and the accommodating groove **141** runs

through two opposite surfaces of the housing **14**. In the embodiment, it may be defined that the accommodating groove **141** runs through the two opposite surfaces of the housing **14** along the first direction X, to define a direction along which the pressing member **541** moves in the accommodating groove **141**, so as to allow the pressing member **541** to move in the accommodating groove **141** along the first direction X. In the embodiment, the accommodating groove **141** may be a notch provided at an edge of the housing **14**, so as to facilitate formation and control over dimensional accuracy of the accommodating groove **141**. For example, in the embodiment illustrated in FIG. **4** and FIG. **5**, the housing **14** may include an outer surface **145**, an inner surface **147** and a side wall surface **149**. The outer surface **145** and the inner surface **147** face away from each other. The outer surface **145** is located on a side of the housing **14** away from the middle frame **12**, the inner surface **147** is located on a side of the housing **14** facing towards the middle frame **12**, and the side wall surface **149** is connected between the outer surface **145** and the inner surface **147**. The accommodating groove **141** runs through the outer surface **145** and the inner surface **147**, and extends to the side wall surface **149**, so as to define a notch structure in the housing **14**, which is beneficial for the formation of the accommodating groove **141**. And when the accommodating groove **141** is configured to accommodate the structure of the key **54**, the structure of the key **54** may be made closer to the edge of the housing **14**, which is beneficial to make more space inside the first structural body **10** available for installation of other electronic components, thereby facilitating improvement of the space utilization. In other embodiments, the accommodating groove **141** may be provided at other portions of the housing **14**. For example, the accommodating groove **141** may be a through hole provided at a position close to an edge of the housing **14**, but it does not need to extend to the side wall surface **149**.

(35) Further, the outer surface **145** of the housing **14** may be a curved surface to provide a smooth appearance surface, so that there is a good touch for the user. When the pressing member **541** is installed in the accommodating groove **141**, the outer surface of the pressing member **541** and the outer surface **145** of the housing **14** may be coplanar, so that the locking structure **100** has a relatively consistent appearance. In the embodiment, the expression “coplanar” may be understood as coplanarity of planar surfaces or/and curved surfaces. That is, the surface of the pressing member **541** and the outer surface **145** of the housing **14** are substantially flush at the joint, where the two are substantially continuous arc surfaces, and they present a substantially smooth transition at the joint.

(36) In some embodiments, the first structural body **10** may be provided with a position-limiting portion **143**. The position-limiting portion **143** is used to be mate with the pressing member **541** to define a position of the pressing member **541** in the accommodating groove **141**, so that the pressing member **541** can be reliably mounted on the housing **14**. It can be seen that, in the embodiment, the accommodating groove **141** and the position-limiting portion **143** together form a main structure of the first mounting portion **101** for mounting of the pressing member **541**. Certainly, in other embodiments, the structure of the first mounting portion **101** is not limited thereto, and it may further include other snap-fit structures and accommodating structures, as long as the positioning and mounting of the pressing member **541** can be implemented, which will not be enumerated herein.

(37) As illustrated in FIG. **6**, the second mounting portion **103** is a mounting structure provided at the middle frame **12**. For example, the second mounting portion **103** may include a hole, a groove, a snap-fit structure or the like on the middle frame **12**. For example, the middle frame **12** may be provided with a positioning structure for mating with the clamped member **543**, and the positioning structure may be regarded as the second mounting portion **103**. For example, the middle frame **12** may be provided with a first positioning portion **105**, and the clamped member **543** may be correspondingly provided with a second positioning portion **5438** (illustrated in FIG. **8**). The second positioning portion **5438** mates with the first positioning portion, to position the clamped member **543** at the middle frame **12**. In this case, the first positioning portion **105** may be regarded

as constituting the structure of the second mounting portion **103**, or it may be considered that the second mounting portion **103** includes the first positioning portion **105**. The structure of the first positioning portion **105** is not limited, and it may be a hole, a groove structure, a snap-fit structure or the like provided on the middle frame **12**. In the embodiment illustrated in FIG. **6**, the second mounting portion **103** or the first positioning portion **105** may include a receiving space **121** provided in the middle frame **12**, and the receiving space **121** is configured to receive the clamped member **543**. The receiving space **121** and the accommodating groove **141** are generally arranged opposite each other, to jointly receive the connecting assembly **50**, and allow the pressing member **541** in the accommodating groove **141** to push the clamped member **543** in the receiving space **121** to move.

(38) In the embodiment, the second mounting portion **103** or the first positioning portion **105** may further include a holding hole **127** provided in the middle frame **12**. The holding hole **127** is communicated with the receiving space **121**, and is configured to hold the second positioning portion **5438** of the clamped member **543**, so as to position the clamped member **543** in the receiving space **121**. For example, the second positioning portion **5438** may be a holding protrusion **5439** (illustrated in FIG. **8**) provided on the clamped member **543**. The holding protrusion **5439** may be movably accommodated in the holding hole **127**, to realize positioning and mounting of the clamped member **543**. The size of the holding protrusion **5439** in the first direction X is smaller than the size of the holding hole **127** in the first direction X, so that the holding protrusion **5439** can move along the first direction X in the holding hole **127**, thereby ensuring the movement degree of freedom of the clamped member **543** in the first direction X. Further, the size of the holding protrusion **5439** in the third direction Z is slightly smaller than or equal to the size of the holding hole **127** in the third direction Z, so as to limit the movement degree of freedom of the clamped member **543** in the third direction Z, thereby preventing the clamped member **543** from moving in the third direction Z, and ensuring reliable positioning of the clamped member **543** in the receiving space **121**. It can be seen that, in the embodiment, the holding hole **127** may serve as a guiding structure, and the holding hole **127** and the receiving space **121** together define a main structure (that is, the first positioning portion **105**) of the second mounting portion **103** for mounting of the clamped member **543**.

(39) In some other embodiments, the holding hole **127** may be replaced by a guide groove structure, and the holding protrusion **5439** may be movably provided in the guide groove structure and can move relative to the first structural body **10** in the first direction X. In other embodiments, a mating structure enabling the positioning between the clamped member **543** and the second mounting portion **103** may include a guide rail and a guide channel which mate with each other. For example, one of the first positioning portion **105** and the second positioning portion **5438** includes a guiding structure such as a guide rail or a guide channel, where such guiding structure is provided in a predetermined direction such as the first direction X; and the other one of the first positioning portion **105** and the second positioning portion **5438** includes a sliding component (for example, a sliding block, a protrusion or a like structure). The sliding component may be slidably connected to the guiding structure such as the guide rail (or the guide channel), and may be slide on the guiding structure along the first direction X. In the way, the clamped member **543** can be conveniently positioned at the first structural body **10** through the second mounting portion **103**.

(40) In other embodiments, the holding hole **127** may be omitted. For example, the second mounting portion **103** or the first positioning portion **105** may include only the receiving space **121** provided in the middle frame **12**, and the size of the receiving space **121** is adapted to the size of the clamped member **543** to position and install the clamped member **543** to the second mounting portion **103**. Specifically, the middle frame may include an inner wall **1210** provided at the receiving space **121**. The inner wall **1210** defines a scope of the receiving space **121** in such a manner that the receiving space **121** has a size parameter which may be referenced relatively, and the clamped member **543** is provided in a space surrounded by the inner wall **1210**. Further, the

size of the holding protrusion **5439** in the first direction X is smaller than the size of the receiving space **121** in the first direction X, so that the holding protrusion **5439** can move along the first direction X in the receiving space **121**, thereby ensuring the movement degree of freedom of the clamped member **543** in the first direction X. Further, the size of the holding protrusion **5439** in the third direction Z is slightly smaller than or equal to the size of the receiving space **121** in the third direction Z, so as to restrict the movement degree of freedom of the clamped member **543** in the third direction Z, thereby preventing the clamped member **543** from moving in the third direction Z, and ensuring reliable positioning of the clamped member **543** in the receiving space **121**.

(41) Certainly, in other embodiments, the structure of the second mounting portion **103** is not limited thereto, and the second mounting portion **103** may further include other snap-fit structures and accommodating structures, as long as the positioning and mounting of the clamped member **543** can be implemented, which will not be enumerated herein. In some embodiments, the second mounting portion **103** may also be considered as at least a part of the structure of the middle frame **12**. For example, the second mounting portion **103** may be understood as a part of the middle frame **12** where the receiving space **121** is provided, or may be understood as a part of the middle frame **12** where the holding hole **127** or a hole or groove structure used for positioning is provided, or may be understood as a part of the middle frame **12** where the receiving space **121** and the holding hole **127** are provided, or may also be understood as the whole structure of the middle frame **12**. On this basis, the clamped member **543** can be conveniently positioned to the first structural body **10** through the second mounting portion **103** (for example, a physical structure of the middle frame **12**, a hole or a groove structure), and the positioning does not depend on the connection or fixing structure between the clamped member **543** and the pressing member **541**. Thus, the positioning and mounting of the clamped member **543** can be performed separately from the positioning and mounting of the pressing member **541**, the original dimension chain of the key **54** is broken, and there is a short dimension chain generated when the key **54** is mounted to the first structural body **10**, which reduces the difficulty of producing and assembling the locking structure **100**.

(42) When the clamped member **543** is mounted to the second mounting portion **103**, at least partial structure of the clamped member is arranged opposite to the pressing member **541** mounted to the first mounting portion **101**. When the pressing member **541** is pressed by an external force, the pressing member **541** moves towards the clamped member **543** along the first direction X, and can push the clamped member **543** to move along the first direction X. On the basis that the positioning and mounting of the pressing member **541** are performed separately from the positioning and mounting of the clamped member **543**, transmission of movement is enabled between the pressing member and the clamped member by means of the opposite position relationship thereof, which avoids the difficulty in installation that is caused due to a too long dimension chain generated when the pressing member **541** and the clamped member **543** are fixedly connected through other fastening structures. In this way, the pressing member **541** is positioned and mounted at the housing **14**, and the clamped member **543** is positioned and mounted at the middle frame **12**, and there is no need to make the pressing member **541** and the clamped member **543** strictly assembled together and then mounted to the first structural body **10**. There is no longer a dimension chain for assembly between the pressing member **541** and the clamped member **543**, and there is a short dimension chain generated when the key **54** is mounted onto the first structural body **10**, which reduces the difficulty of producing and assembling the locking structure **100**.

(43) With regard to the structure that the pressing member **541** and the clamped member **543** are arranged opposite to each other after being mounted to the first structural body **10**, it is not limited to one that the pressing member and the clamped member are directly opposite to and spaced apart from each other, and a middle element or component may be provided therebetween to improve the stability of the overall structure of the connecting assembly **50**. For example, in some embodiments, as illustrated in FIG. **7**, the key **54** may further include a buffer component **545**, and

the buffer component **545** is provided between the pressing member **541** and the clamped member **543**. The buffer component **545** may be made of an elastic material, and may be configured to reduce an impact force between the pressing member **541** and the clamped member **543**, thereby avoiding wear to the mating parts between the pressing member **541** and the clamped member **543**, and improving the hand feeling of pressing the key **54**. Further, the buffer component **545** may be configured to apply a supporting force to the pressing member **541**, which enables the mating between the pressing member **541** and the position-limiting portion **143** to be reliable. Thus, the size of a gap between the pressing member **541** and the side wall of the accommodating groove **141** is enabled to be in the preset range. This avoids a position offset of the key **54** relative to the accommodating groove **141** that would be caused due to loosening of the key **54** after being reused many times. In addition, this ensures that the circumferential mounting gap of the key **54** always meets a predetermined dimensional requirement, where the mounting gap is uniform and controllable. Accordingly, the structure of the locking structure **100** is enabled to have a high quality, and it is beneficial to ensure a good use experience of the user.
(44) As illustrated in FIG. **7**, in the embodiment, when the second structural body **30** is confined at the first structural body **10**, there is a spacing **547** between the pressing member **541** and the clamped member **543** in the moving direction along which the pressing member **541** moves relative to the first structural body **10**, and at least part of the buffer component **545** is provided in the spacing **547** to fill the spacing **547**. In the embodiments of the present disclosure, one object being confined at another object may be understood that one object may limit the movement range of the other object, and the two objects are generally difficult to be disengaged from each other. For example, in the embodiment, the second structural body **30** being confined at the first structural body **10** means that, the first structural body **10** may limit the movement range of the second structural body **30**, and the second structural body **30** generally cannot easily be detached from the first structural body **10**. When the second structural body **30** is confined at the first structural body **10**, due to a gap between the second structural body **30** and the first structural body **10**, the second structural body **30** can move or rotate relative to the first structural body **10** within a small range. In some embodiments, the second structural body **30** may also be in tight fit with the first structural body **10** such that the second structural body **30** is almost fixedly connected to the first structural body **10**.
(45) In some embodiments, the pressing member **541** has a first position and a second position relative to the first structural body **10**. In the first position, the clamped member **543** is configured to: mate with the second structural body **30**, so as to make the second structural body **30** confined at the first structural body **10**; and squeeze the buffer component **545**, to make the buffer component **545** abut against the pressing member **541**, so as to make the pressing member **541** confined at the groove **1031**. When being pressed, the pressing member **541** moves to the second position relative to the first structural body **10**, and in the process of switching the pressing member **541** from the first position to the second position, the pressing member **541** squeeze the buffer component **545** and drives the clamped member **543** to move. In the second position, the clamped member **543** is disengaged from the second structural body **30**, so that the second structural body **30** can be detached from the first structural body **10**.
(46) Further, in the embodiment, the buffer component **545** may be a block made of an elastic material, so as to improve the structural stability thereof, and enhance the supporting force to the pressing member **541**. For example, the buffer component **545** may be made of a material such as rubber or plastic. The buffer component **545** may also be an elastic structure having an elastic deformation capability, for example, it may be an elastic structure made of a non-elastic material. Specifically, the buffer component **545** may be an elastic structure such as an elastic sleeve, an elastic sheet, or a spring, to apply a supporting force to the pressing member **541**, so that the size of the gap between the pressing portion **5411** and the side wall of the accommodating groove **141** meets the requirement, and transmission of movement may be reliably performed between the

pressing member **541** and the clamped member **543**. In the embodiments of the present disclosure, since the buffer component **545** capable of generating elastic deformation is provided between the pressing member **541** and the clamped member **543**, the elastic deformation of the buffer component **545** can be utilized to adapt to the change in the size of the gap between the pressing member **51** and the clamped member **543**, thereby making the assembly gap between the components of the locking structure **100** more easily meet the tolerance requirement.
(47) For example, in a structure in which the buffer component **545** is not used, if the pressing member **541** and the clamped member **543** are in direct contact with each other, when any one of the pressing member **541** and the clamped member **543** has a manufacturing tolerance, the assembly obtained by directly making the pressing member **541** abut against the clamped member **543** has a long dimension chain, which would inevitably cause a large installation error. When the pressing member **541** and the clamped member **543** need to be mounted onto the first structural body **10** after being assembled together, it would be unable to perform the mounting since the large installation error would probably cause interference with other parts of the first structural body **10**. In this case, in order to reduce the probability of this phenomenon, there are high requirements on the manufacturing accuracy of the pressing member **541** and the clamped member **543**, and the production cost is also increased. In the present embodiment, the pressing member **541** and the clamped member **543** are arranged opposite to and spaced apart from each other and a mounting gap is provided therebetween, and the buffer component **545** is provided in the mounting gap between the pressing member **541** and the clamped member **543**. As such, when any one of the pressing member **541** and the clamped member **543** has a manufacturing tolerance, in order to limit the installation error produced after the pressing member **541**, the clamped member **543** and the buffer component **545** are assembled, the buffer component **545** may be compressed to make the mounting gap reduced. In this case, at the time of mounting the pressing member **541**, the clamped member **543** and the buffer component **545**, which have been assembled together, to the first structural body **10**, the mounting gap can be changed due to the elastic deformation (compression) of the buffer component **545**, so that the overall structure and size of the three are adapted to the size requirement of the first structural body **10**. This means that the manufacturing tolerances and the installation errors of the pressing member **541** and the clamped member **543** are compensated by the buffer component **545**, which greatly reduces the requirements on the manufacturing accuracy of the pressing member **541** and the clamped member **543**, and also reduces the assembly difficulty and the production cost.
(48) In the embodiments of the present disclosure, when the buffer component **545** is provided between the pressing member **541** and the clamped member **543**, the connection of the buffer component **545** with each of the pressing member **541** and the clamped member **543** is not limited. In some embodiments, the buffer component **545** may be directly provided between the pressing member **541** and the clamped member **543**, without being physically connected with the pressing member **541** or the clamped member **543**. For example, the buffer component **545** may be positioned by means of the structure of the middle frame **12**, or the buffer component **545** is clamped by the pressing member **541** and the clamped member **543**. In other embodiments, the buffer component **545** may be physically connected with one of the pressing member **541** and the clamped member **543**, and be arranged opposite to and spaced apart from or overlapped with or attached to the other one of the pressing member **541** and the clamped member **543**. In other embodiments, two sides of the buffer component **545** may be connected to the pressing member **541** and the clamped member **543**, respectively; for example, the buffer component **545** may be connected to each of the pressing member **541** and the clamped member **543** through a mating structure including a groove and a protrusion, this is beneficial for the buffer component **545**, the pressing member **541** and the clamped member **543** to substantially form a whole module, the structure of which is stable and reliable, and which is convenient to disassemble and assemble.
(49) As illustrated in FIG. **7** and FIG. **8**, in the embodiments, the buffer component **545** is

connected to the clamped member **543**. Further, the buffer component **545** is provided with a first connection portion **5451**. The first connection portion **5451** is configured to be detachably connected with the clamped member **543**, so that the structure of the buffer component **545** is more stable and reliable. In other embodiments, the first connection portion **5451** may be configured to be connected to the pressing member **541**.
(50) In this embodiment, in order to adapt to the connection structure of the buffer component **545**, the clamped member **543** may include an abutting portion **5431** and a clamped portion **5433**. The abutting portion **5431** is configured for connection with the buffer component **545**. The clamped portion **5433** is connected to the abutting portion **5431**, and is configured to be engaged with the locking portion **34**.
(51) The abutting portion **5431** is provided on a side facing towards the pressing member **541**, and may be arranged opposite to and spaced apart from the pressing member **541**, or may be in a surface contact with the pressing member **541**, or may be in contact with the pressing member **541** through a middle component (such as the buffer component **545**). For example, in the embodiment, a buffer component **545** is provided between the clamped member **543** and the pressing member **541**, the abutting portion **5431** abuts against the buffer component **545**, and the buffer component **545** abuts against an end of the pressing member **541**. The clamped portion **5433** is provided at an end of the abutting portion **5431**, and is arranged opposite to and spaced apart from the first structural body **10**. A clamping space **5434** is defined between the clamped portion **5433** and the first structural body **10**. The clamping space **5434** is configured to receive the locking portion **34** of the second structural body **30**; at that time, the locking portion **34** is movably provided in the clamping space **5434**.
(52) Further, as illustrated in FIG. **9**, in the embodiment, the locking structure **100** may include a baffle **16** detachably connected to the first structural body **10**. The baffle **16** is provided at the second mounting portion **103**, and shields at least part of the clamped member **543**. The baffle **16** is provided with a through hole **161** for receiving the locking portion **34**. The cross section of the baffle **16** may be racetrack shaped, and may be provided with a connection hole and the through hole **161** corresponding to the locking portion **34**. The connection hole may be configured to receive a threaded fastener, to achieve the detachable connection of the baffle **16** to the first structural body **10**. Further, the baffle **16** may also limit the movement of the clamped member **543** in the second mounting portion **103**, to enable the clamped member **543** to move smoothly in the second mounting portion **103**.
(53) After the locking portion **34** is mounted to the first structural body **10**, the main body **32** is located on a side of the baffle **16** facing away from the clamped member **543**, and the through hole **161** can restrict movement of the locking portion **34** in a moving direction relative to the first structural body **10**. In an embodiment in which there are two locking portions **34**, the number of the through holes **161** is the same as the number of the locking portions **34**, and the shape of the through hole **161** matches the shape of the locking portion **34**. For example, in some embodiments, the locking portion **34** is cylindrical, the through hole **161** is circular, and the diameter of the circular hole is substantially equal to the diameter of the locking portion **34**. In some other embodiments, the locking portion **34** may be prismatic, such as a triangular prism, a quadrangular prism, or a pentagonal prism, and the shape of the cross section of the through hole **161** matches the shape of the locking portion **34**. As such, the through hole **161** and the locking portion **34** may fit tightly, which facilitates the waterproof and dustproof design of the wearable device. With the through hole **161**, the movement of the locking portion **34** may also be guided and limited, which prevents the movement of the locking portion **34** in the second mounting portion **103** from being deflected, and enables the locking portion **34** to smoothly push the clamped member **543** to move in the second mounting portion **103**.
(54) In order to ensure the reliability of the abutting between the abutting portion **5431**, the buffer component **545** and the pressing member **541**, the connecting assembly **50** may further include a

position-restoring member **58**. The position-restoring member **58** is provided on a side of the abutting portion **5431** away from the buffer component **545**, and located between the abutting portion **5431** and the first structural body **10** (for example, the middle frame **12** or the housing **14**), to apply, to the abutting portion **5431**, a supporting force towards the buffer component **545**. In the process of detaching the second structural body **30** from the first structural body **10**, the pressing member **541** may be pressed, and then the position-restoring member **58** is compressed to apply, to the clamped member **543**, a supporting force towards the buffer component **545** and the pressing member **541**. When the external force applied to the pressing portion **5411** is withdrawn, the position-restoring member **58** may apply a restoring force that provides movement of the whole of the clamped member **543** and the pressing member **541**. In some embodiments, in an ordinary state, the position-restoring member **58** further applies, to the clamped member **543**, a supporting force towards the buffer component **545** and the pressing member **541** (for example, in the ordinary state, the position-restoring member **58** is located between the clamped member **543** and the first structural body **10** and is in a compressed state), to ensure the reliability of the connection between the clamped member **543** and the locking portion **34**.

(55) Further, a side of the abutting portion **5431** facing away from the second positioning portion **5438** is provided with a receiving groove **5432**, and at an end of the abutting portion **5431** facing away from the pressing member **541**, the receiving groove **5432** extends to an end surface of the abutting portion **5431**, and one end of the position-restoring member **58** is accommodated in the receiving groove **5432**. In the present embodiment, there are two position-restoring members **58**, and there is a one-to-one correspondence between the position-restoring members **58** and the receiving grooves **5432**. The second positioning portion **5438** may guide and limit the movement of the clamped member **543** in the accommodating groove **141**. In some other embodiments, the position of the receiving groove **5432** may correspond to the position of the second positioning portion **5438**, that is, the thickness of the abutting portion **5431** is thicker at the position where the receiving groove **5432** is located. Such arrangement can not only use the mating of the second positioning portion **5438** with the accommodating groove **141** to guide and limit the clamped member **543**, but also improve the structural strength of the abutting portion **5431** at the receiving groove **5432**, to improve the strength of the connection structure between the position-restoring member **58** and the abutting portion **5431**.

(56) As illustrated in FIG. **10**, in other embodiments, the position-restoring member **58** may include a first magnet **581** and a second magnet **583**. The first magnet **581** is connected to the clamped member **543**, and the second magnet **583** is connected to the first structural body **10**. In the process of mounting the locking portion **34** into the second mounting portion **103**, the first magnet **581** and the second magnet **583** approach each other and generate a magnetic repulsion force.

(57) Further, as illustrated in FIG. **8**, in order to enhance the connection between the buffer component **545** and the clamped member **543**, in some embodiments, the clamped member **543** may further include a second connection portion **5435**. The second connection portion **5435** is connected to the abutting portion **5431**, and is configured to mate with the first connection portion **5451** of the buffer component **545**, such that the buffer component **545** is fixed relative to the clamped member **543**. One of the first connection portion **5451** and the second connection portion **5435** is a groove, and the other one there is a protrusion matching the groove. Though engagement of the groove and the protrusion, the second connection portion **5435** and the first connection portion **5451** may mate with each other by means of a simple structure. Specifically, the first connection portion **5451** may be a groove provided in the buffer component **545**, and the second connection portion **5435** may be a protrusion provided on the abutting portion **5431**. In other embodiments, the first connection portion **5451** and the second connection portion **5435** may be other connection structures, such as a hook structure or a sticker structure. In the embodiment, the connection structure between the first connection portion **5451** and the second connection portion **5435** is applied between the buffer component **545** and the pressing member **541**. It is

understandable that, in other embodiments, the connection structure between the first connection portion **5451** and the second connection portion **5435** may be applied between the buffer component **545** and the pressing member **541**. For example, the second connection portion **5435** may be provided at one end of the pressing member **541**, and mate with the first connection portion **5451** of the buffer component **545**, which will not be enumerated herein.
(58) In the embodiment illustrated in FIG. **8**, the pressing member **541** is movably provided in the accommodating groove **141**, to enable the key **54** to move in the accommodating groove **141** along the first direction X. The pressing member **541** includes a pressing portion **5411** and a projecting portion **5413** connected to the pressing portion **5411**. The projecting portion **5413** may be integrally formed with the pressing portion **5411**. In the embodiment, the pressing portion **5411** is substantially block-shaped, and is movably accommodated in the accommodating groove **141**. The surface of the pressing portion **5411** is substantially coplanar with the outer surface **145** of the housing **14**, such that the locking structure **100** has a consistent appearance. The projecting portion **5413** is provided on a side of the pressing portion **5411** away from the housing **14**, projects out relative to the pressing portion **5411**, and extends towards the clamped member **543**. The projecting portion **5413** is substantially cylindrical, for connection to the clamped member **543**.
(59) In some embodiments, an elastic connection member **52** is fixedly connected to at least one of the pressing portion **5411** and the projecting portion **5413**. The elastic connection member **52** is provided in the key groove **1011**. And in the first position, the elastic connection member **52** abuts against the housing **14**, such that the end surface of the pressing portion **5411** exposed from the housing **14** is flush with the surface of the housing **14** at the key groove **1011**.
(60) In some embodiments, the elastic connection member **52**, the pressing portion **5411** and the projecting portion **5413** are integrally formed. For example, the elastic connection member **52** may be a projection extending from an edge of the pressing portion **5411**. When the pressing member **541** is in the first position, the elastic connection member **52** abuts against a groove wall of the key groove **1011**, so that the position of the pressing member **541** at the housing **14** is relatively fixed, and the end surface of the pressing portion **5411** exposed from the housing **14** is flush with the end surface of the housing **14** at the key groove **1011**. It is understandable that, only the machining accuracy of the key groove **1011** and the machining accuracy of the elastic connection member **52** meet the requirements of assembly accuracy, the end surface of the pressing portion **5411** exposed from the housing **14** and the end surface of the housing **14** at the key groove **1011** can match at a high precision, to meet the requirement of being flush. For example, the elastic connection member **52** of the pressing member **541** may use the housing **14** as an assembly reference, to shorten the dimension chain for assembly, and decrease the influence of machining and assembly tolerances, so that it is easy to enable the end surface of the pressing portion **5411** exposed from the housing **14** to be flush with the end surface of the housing **14** at the key groove **1011**.
(61) In the present embodiment, the projecting portion **5413** is provided between the pressing portion **5411** and the clamped member **543**. When an external force is applied to press the pressing portion **5411**, the pressing portion **5411** drives the projecting portion **5413** to push the clamped member **543** to move along the first direction X, so as to release the engagement between the clamped member **543** and the locking portion **34**. In the process of mounting the second structural body **30** to the first structural body **10**, the locking portion **34** may move along the second direction Y to be inserted into the first structural body **10**, and the clamped member **543** is driven by the locking portion **34** to move relative to the first structural body **10**, and is restored under the support of the position-restoring member **58** so as to be engaged with the locking portion **34**, where the second direction Y may substantially intersect with the first direction X (for example, the second direction Y is substantially perpendicular to the first direction X). During the process of detaching the second structural body **30** from the first structural body **10**, the pressing member **541** can be pressed to enable the clamped member **543** to be disengaged from the locking portion **34**.
(62) In order to effectively position the pressing portion **5411**, a positioning connection member

mating with the position-limiting portion **143** may be provided to enable the pressing portion **5411** to be positioned in the housing **14** of the first structural body **10**. For example, as illustrated in FIG. **11**, in some embodiments, the connecting assembly **50** may further include an elastic connection member **52**. The elastic connection member **52** is connected to the pressing portion **5411**, and is configured to mate with the position-limiting portion **143** of the housing **14** to position the pressing member **541**. The elastic connection member **52** is provided in the first mounting portion **101**, and in the first position, the elastic connection member **52** limits the movement range of the pressing member **541**. Further, as illustrated in FIG. **11** and FIG. **12**, the elastic connection member **52** is provided with a positioning portion **5201**, and the positioning portion **5201** is configured to mate with the position-limiting portion **143**, to enable the pressing member **541** to perform limited movement (for example, move in the first direction X) in the accommodating groove **141**. In the process of mounting the pressing member **541** to the accommodating groove **141**, the elastic connection member **52** is capable of being squeezed and deformed by the side wall of the accommodating groove **141**, until the positioning portion **5201** is accommodated in the position-limiting portion **143** to mate therewith, so as to define the position of the pressing member **541** relative to the first structural body **10**.
(63) In the embodiments of the present disclosure, the deformation form of the elastic connection member **52** is not limited. For example, the elastic connection member **52** itself may be elastically deformed (for example, the elastic connection member **52** itself is an elastic structural body or made of an elastic material), so that the positioning portion **5201** may deform or/and extend-retract relative to the pressing member **541**, to facilitate disassembly and assembly. For another example, the elastic connection member **52** may include an elastic structural body, and through deformation of the elastic structural body, the positioning portion **5201** can deform or/and extend-retract relative to the pressing member **541**, to facilitate disassembly and assembly. In the embodiment, the elastic connection member **52** itself is in a telescopic structure, and through the deformation generated by the extending or retraction of the telescopic structure, the positioning portion **5201** can extend or retract relative to the pressing member **541** to facilitate disassembly and assembly.
(64) In the embodiment illustrated in FIG. **11** and FIG. **12**, the elastic connection member **52** includes a fixing portion **521**. The fixing portion **521** is fixedly connected to the pressing portion **5411** of the pressing member **541** (for example, the pressing portion **5411** may be provided with a receiving groove for receiving the fixing portion **521**). The positioning portion **5201** is movably connected to the fixing portion **521**, and may protrude relative to the pressing portion **5411**. During the process of mounting the pressing member **541** to the accommodating groove **141**, the positioning portion **5201** can be squeezed by the side wall of the accommodating groove **141** to retract relative to the fixing portion **521**, and can extend out relative to the fixing portion **521** to mate with the position-limiting portion **143**, to define the position of the pressing member **541** relative to the first structural body **10**. Further, the positioning portion **5201** is movably accommodated in the position-limiting portion **143** of the housing **14**, in such a manner that the key **54** can move in the accommodating groove **141** along the first direction X. In the embodiment, the positioning portion **5201** is a protrusion structure, and the position-limiting portion **143** mating with the positioning portion **5201** is a groove structure, and the two are hold together through engagement. It is understandable that, the expressions “hold”, “engage” or “clamp” in the present disclosure should be understood as that two parties mating with each other may be provided with corresponding receiving and inserting structures (such as the groove structure and the protrusion structure engaged with each other as mentioned above), or a clamping structure, etc., and it does mean that the two parties mating with each other need to get stuck to each other by means of an external force.
(65) As illustrated in FIG. **12** and FIG. **13**, in the present embodiment, the fixing portion **521** is substantially rod-shaped, and is provided with an accommodating cavity **5211** for accommodating the positioning portion **5201**. The accommodating cavity **5211** may be provided along a length

direction of the fixing portion **521**, and run through two opposite ends of the fixing portion **521**. There may be two positioning portions **5201**, and the two positioning portions **5201** may be provided at two opposite ends of the accommodating cavity **5211** respectively, and both of them may extend out of the fixing portion **521** through the accommodating cavity **5211**, to protrude relative to both ends of the pressing portion **5411** respectively, thereby mating with the two position-limiting portions **143**. Further, the elastic connection member **52** may further include an elastic portion **523** provided between the two positioning portions **5201**, to provide, to the positioning portion **5201**, a supporting force that enables the positioning portion **5201** to extend out of the fixing portion **521**. Further, a position-limiting boss **5213** is provided within in the accommodating cavity **5211** of the fixing portion **521**, and the position-limiting boss **5213** may be provided at an end position of the accommodating cavity **5211**. A position-limiting flange **5215** may be provided at the positioning portion **5201**. The position-limiting flange **5215** is accommodated in the accommodating cavity **5211**, and is arranged opposite to the position-limiting boss **5213**. When the position-limiting flange **5215** abuts against the position-limiting boss **5213**, the position-positioning portion **5201** arrives at its extreme position for extending out from the fixed portion **521**, to prevent the positioning portion **5201** from falling off the fixing portion **521**.
(66) Since the first structural body **10** is the main supporting structure for mounting the connecting assembly **50**, other details of the first structural body **10** will be described below according to some embodiments of the present disclosure.
(67) As illustrated in FIG. **13** and FIG. **14**, the middle frame **12** includes a main body portion **120**, and the main body portion **120** forms a main supporting structure of the first structural body **10**. The main body portion **120** may be substantially in a shape of a rectangular frame, a round frame, or an oval frame, or in other irregular geometric shapes or irregular shapes, which is not limited in the present disclosure. The main body portion **120** includes a bearing surface **123** and a side surface **125** connected to the bearing surface **123**, where the bearing surface **123** is configured to bear the housing **14**. In this embodiment, an orientation of the bearing surface **123** is different from an orientation of the side surface **125**. Further, the bearing surface **123** is substantially perpendicular to the first direction X, and the side surface **125** is substantially perpendicular to the second direction Y. The bearing surface **123** and the side surface **125** may be planar or curved surfaces, or may be planar or curved surfaces with grooves or/and protrusions, which is not limited in the present disclosure. The receiving space **121** of the middle frame **12** is provided inside the main body portion **120**, and runs through the bearing surface **123** and the side surface **125**. The receiving space **121** runs through the bearing surface **123** to define a first opening **1211**, and the first opening **1211** is configured to mount and accommodate the key **54**. The receiving space **121** runs through the side surface **125** to define a second opening **1213**, and the second opening **1213** is configured to mount the locking portion **34** of the second structural body **30**. For example, in the process of mounting the second structural body **30** to the first structural body **10**, the locking portion **34** may pass through the second opening **1213** so as to be engaged with the clamped member **543** in the receiving space **121**. In the embodiment, the orientation of the first opening **1211** is substantially consistent with the first direction X, and the orientation of the second opening **1213** is substantially consistent with the second direction Y, for example it is substantially perpendicular to the orientation of the first opening **1211**.
(68) As illustrated in FIG. **14** and FIG. **15**, in some embodiments, the middle frame **12** may further include a boundary portion **129**. The boundary portion **129** is provided on a side of the main body portion **120**, and is configured to cover the main body portion **120**, and define an appearance surface of the locking structure **100** together with the housing **14**. In the embodiment, the boundary portion **129** may be covered on the side surface **125** of the main body portion **120**, and partially cover the second opening **1213** so as to prevent too much exposure of the second opening **1213** to the external environment; this is beneficial to enable an excellent dustproof performance of the locking structure **100**, and can also reduce the number or/and area of the openings provided in the

appearance surface of the locking structure **100**, so as to improve the consistency of the appearance structure. Further, the boundary portion **129** may be provided with a through hole **1291** communicated with the second opening **1213**, where the size of the through hole **1291** is smaller than the size of the second opening **1213**, and the through hole **1291** is configured to allow the locking portion **34** of the second structural body **30** to pass therethrough. Further, when the boundary portion **129** is covered the side surface **125** of the middle frame **12**, at least part of the boundary portion **129** is arranged opposite to the side wall surface **149** of the housing **14**, so that the boundary portion **129** may cover an opening part of the notch structure of the accommodating groove **141**, thereby limiting the degree of freedom of movement of the key **54** in the second direction Y, thereby preventing the key **54** from falling off from the opening part, where the second direction Y may intersect with the first direction X (for example, the two are substantially perpendicular to each other).

(69) Further, in the embodiment, when the boundary portion **129** is provided on a side of the side surface **125**, the boundary portion **129** may protrude relative to the bearing surface **123**, so as to be arranged opposite to the side wall surface **149** of the housing **14**. As such, when the housing **14** is covered on the bearing surface **123**, the protruding part of the boundary portion **129** and the side wall surface **149** which are arranged opposite to each other may enable the middle frame **12** and the housing **14** to be tightly combined with each other, thereby ensuring an excellent sealing effect enabling waterproof and dustproof of the first structural body **10**. In some embodiments, the material of the boundary portion **129** may be different from the material of the main body portion **120**. For example, the boundary portion **129** and the main body portion **120** may be produced through processing such as double color injection or insert molding, and connected with each other. The boundary portion **129** may substantially cover the side surface **125**, or may be connected to only one side of the side surface **125** or connected to only the bearing surface **123**, and protrude relative to the bearing surface **123**, which is not limited in the present specification.

(70) When the housing **14** is covered on the bearing surface **123**, the accommodating groove **141** is arranged substantially opposite to the first opening **1211** of the middle frame **12**.

(71) As illustrated in FIG. **14** and FIG. **15**, the position-limiting portion **143** is provided adjacent to the accommodating groove **141**, and is configured to define the position of the elastic connection member **52**, so as to define a relative position of the pressing member **541** in the accommodating groove **141**. In the embodiment illustrated in FIG. **14** and FIG. **15**, the position-limiting portion **143** is generally a groove structure, and is configured to accommodate the positioning portion **5201** of the elastic connection member **52**. The specific arrangement position of the position-limiting portion **143** is not limited, and in the embodiment illustrated in FIG. **14** and FIG. **15**, the position-limiting portion **143** may be provided on a side wall of the accommodating groove **141**, and extend to a surface of the housing **14** facing towards the middle frame **12**. There may be two position-limiting portions **143**, and the two position-limiting portions **143** may be respectively provided at two opposite ends of the receiving groove **141**. In other embodiments, the position-limiting portion **143** may be provided at other positions. For example, the position-limiting portion **143** may be provided on an inner wall of the accommodating groove **141** without extending to the inner surface **147** of the housing **14**, and in this case, the position-limiting portion **143** may be a structure such as a hole or a groove in an inner wall of the accommodating groove **141**. For another example, the position-limiting portion **143** may be provided at the inner surface **147** of the housing **14**, without being communicated with the accommodating groove **141**; and in this case, the position-limiting portion **143** may be a structure such as a hole or a groove provided in the inner surface **147**, and correspondingly, the elastic connection member **52** is also provided at an end of the pressing member **541**. For example, the positioning portion **5201** of the elastic connection member **52** may protrude relative to the pressing portion **5411** and be bent, so as to mate with the position-limiting portion **143**.

(72) In some embodiments, as illustrated in FIG. **8** and FIG. **11** again, the housing **14** may further

include a projection **148**. The projection **148** is provided in the accommodating groove **141**, and projects out relative to a side wall of the accommodating groove **141**. The projection **148** may be provided on a side of the accommodating groove **141** close to the middle frame **12**, and arranged opposite to at least part of the key **54**, to define a position to which the key **54** moves in the accommodating groove **141** along the first direction. Specifically, if the position-limiting portion **143** is a groove and the groove has a bottom wall **1431** opposite to the middle frame **12** (see FIG. **12**), when the pressing member **541** is provided in the receiving groove **141** and the positioning portion **5201** is accommodated in the groove of the position-limiting portion **143**, the positioning portion **5201** is arranged opposite to the bottom wall **1431**, and the bottom wall **1431** of the groove is configured to define a position to which the positioning portion **5201** and the pressing member **541** move in an opposite direction to the first direction X. The projection **148** is arranged opposite to at least part of the pressing member **541**, to define a position to which the pressing member **541** moves in the first direction X. That is, the projection **148** and the bottom wall **1431** of the groove limit the range of movement of the pressing member **541** in the first direction X, which is beneficial to ensure a strong structural stability of the locking structure **100**.
(73) As illustrated in FIG. **15** again, the second structural body **30** is detachably connected to the first structural body **10**. The main body **32** of the second structural body **30** may be provided on a part to which the main body **32** needs to be connected, or may be an integral part of this part. For example, when the locking structure **100** is applied to a smart watch, the main body **32** may be provided on a watch strap of the smart watch, or may be a part of the structure of the watch strap.
(74) As illustrated in FIG. **16**, the locking portion **34** of the second structural body **30** is provided at an end of the main body **32**, and protrudes relative to a surface of the main body **32**. When the second structural body **30** is connected to the first structural body **10**, the locking portion **34** is located on a side of the main body **32** facing towards the first structural body **10**. The locking portion **34** is substantially hook shaped, for mating with the clamped member **543**. Specifically, in the embodiment illustrated in FIG. **16**, the locking portion **34** is provided with a locking groove **341**. The locking groove **341** is configured to receive a part of the structure of the clamped member **543**, to enable clamping between the locking portion **34** and the clamped member **543**. Further, the locking portion **34** may be a columnar structure provided on a surface of the main body **32**, and the locking groove **341** is provided on a peripheral wall of the columnar structure. A side of the locking portion **34** away from the main body **32** may further be provided with a first guiding slope **343**, and the first guiding slope **343** and the locking groove **341** are located on the same side of the column structure. The slope direction of the first guiding slope **343** intersects with an axis direction of the columnar structure, and is used to guide a movement direction of the locking portion **34** at the time of inserting the locking portion **34** into the second opening **1213**, and guide the locking portion **34** to be engaged with the clamped member **543**.
(75) In the embodiments of the disclosure, the number of the locking portions **34** is not limited, and there may be one or more locking portions; and correspondingly, there may also be one or more clamped portions **5433** for the clamped member **543** that are in a one-to-one correspondence to the locking portions **34**, which is not limited in the present disclosure. In the embodiment illustrated in FIG. **14**, FIG. **15** and FIG. **16**, there are two locking portions **34**, and the two locking portions **34** are arranged on the main body **32** in such a manner that they are spaced apart from each other, to enhance the stability of the clamping structure between the second structural body **30** and the clamped member **543**. Correspondingly, there may also be two clamped portions **5433** for the clamped member **543**, and the two clamped portions **5433** are respectively provided at two opposite ends of the abutting portion **5431**, and are provided in one-to-one correspondence with the two locking portions **34**. When the locking portion **34** is inserted into the clamping space **5434**, the clamped portion **5433** is at least partially accommodated in the locking groove **341**, so that the second structural body **30** is connected to the first structural body **10**. When the second structural body **30** needs to be detached, an external force is applied on the pressing member **541** in the first

direction X, and the pressing member **541** pushes the clamped portion **5433** to move along the first direction X so as to be disengaged from the locking groove **341**, and then the second structural body **30** is pulled out so as to be separated from the first structural body **10**; at that time, the pressing member **541** and the clamped member **543** of the key **54** return to their original positions under the action of the restoring member **58**. When the second structural body **30** needs to be connected to the first structural body **10**, the locking portion **34** is inserted along the second direction Y into the clamping space **5434** from the second opening **1213**, and the first guiding slope **343** of the locking portion **34** abuts against the clamped portion **5433** and gradually goes into the clamping space **5434**; in addition, the clamped portion **5433** moves along the first direction X under the pushing of the locking portion **34** to cause the position-restoring member **58** to generate compression deformation, and when the clamped portion **5433** moves to a position where the clamping groove **341** is directly opposite to the clamped portion **5433**, the clamped portion **5433** is pushed into the locking groove **341** under the action of the elastic restoring force of the position-restoring member **58**, at that time, the clamped portion **5433** and the locking portion **34** are engaged. In the above detachment and installation processes, the second structural body **30** can be simply and conveniently mounted and detached. Further, in some embodiments, the clamped portion **5433** may also be provided with a second guiding slope **5437** corresponding to the locking portion **34**, and the second guiding slope **5437** may be provided on a side of the clamped portion **5433** facing towards the main body **32** of the second structural body **30**. For example, in the process of mounting the second structural body **30** to the first structural body **10**, the locking portion **34** abuts against the guiding slope of the clamped member **543** to drive the clamped member **543** to move relative to the second mounting portion **103**. The second guiding slope **5437** is configured to mate with the first guiding slope **343**, to improve fluency of the movement of inserting the locking portion **34** into the clamping space **5434**. In some embodiments, at least one of the clamped portion **5433** and the locking portion **34** is provided with a guiding slope.
(76) In the embodiment, a first transition surface **345** is provided between the first guiding slope **343** and the locking groove **341** of the locking portion **34**, and the first transition surface **345** may be a planar or cambered surface. In the process of mounting the locking portion **34** of the second structural body **30** into the second mounting portion **103**, after the first guiding slope **343** drives the clamped member **543** to move and it moves to the position of the clamped member **543**, the locking portion **34** may continue moving a certain distance towards the inside of the locking structure **100**. In this process, the first guiding slope **343** leaves the second guiding slope **5437**, and the clamped portion **5433** of the clamped member **543** abuts against the first transition surface **345** and the first transition surface **345** moves relative to the clamped portion **5433**, until the clamped portion **5433** is separated from the first transition surface **345** and enters the locking groove **341** of the locking portion **34**; in addition, the clamped member **543** may be driven to move along a direction towards the pressing member **541**, and the locking groove **341** may be utilized to limit the position of the clamped portion **5433**, to prevent the second structural body **30** from falling out of the first structural body **10**. For example, in the embodiment, a surface of the locking groove **341** facing away from the first guiding slope **343** is planar, and is substantially perpendicular to the movement direction of the locking portion **34**. For example, the surface of the locking groove **341** facing away from the first guiding slope **343** or the surface of the locking groove **341** facing towards the main body **32** may be planar, and defines an included angle of 85 degrees to 90 degrees with regard to the movement direction of the latching portion **34**. Thus, the position of the locking portion **34** can be effectively limited by the clamped portion **5433**, so as to prevent the second structural body **30** from being easily disengaged from the first structural body **10** when the second structural body **30** is pulled.
(77) Certainly, a surface of the clamped portion **5433** facing away from the second guiding slope **5437** may also be planar, and is substantially perpendicular to the movement direction of the locking portion **34**. For example, the surface of the clamped portion **5433** facing away from the

second guiding slope **5437** may also be planar, and defines an included angle of 85 degrees to 90 degrees with regard to the movement direction of the locking portion **34**. Thus, the position of the locking portion **34** can be effectively limited by the clamped portion **5433**, so as to prevent the second structural body **30** from being easily disengaged from the first structural body **10** when the second structural body **30** is pulled.

(78) A side of the clamped portion **5433** facing towards the pressing member **541** may be provided with a second transition surface **5436**, and the second transition surface **5436** may be a planar or cambered surface. During the process of mounting the second structural body **30** into the first structural body **10**, after the first guiding slope **343** leaves the second guiding slope **5437**, the locking portion **34** may abut against the second transition surface **5436** and continue moving towards the inside of the electronic device **500** along the second transition surface **5436**, until the locking portion **34** leaves the second transition surface **5436**. For example, in the embodiment in which the locking portion **34** is provided with the first transition surface **345**, after the first transition surface **345** leaves the second transition surface **5436**, the clamped member **543** may enter the locking groove **341** to enable the locking portion to mate with the clamped portion **5433**, that is, to enable the second structural body **30** to be confined at the first structural body **10**.

(79) The first transition surface **345** may enhance the structural strength of the locking portion **34** at the edge of the locking groove **341**. The second transition surface **5436** may enhance the strength of the clamped portion **5433** at the edge of the second guiding slope **5437**. It is understandable that the first transition surface **345** and the second transition surface **5436** are not necessary. For example, in a case where both the first transition surface **345** and the second transition surface **5436** are absent, after the first guiding slope **343** leaves the second guiding slope **5437**, the clamped member **543** may enter the locking groove **341** of the locking portion **34**, and mate with the locking portion **34** for position limiting. In other words, at least one of the first transition surface **345** and the second transition surface **5436** may be absent.

(80) It is understandable that, in the above embodiments, the first position and the second position may be regarded as extreme positions of the pressing member **541** during movement, and a third position and a fourth position may be regarded as extreme positions of the clamped member **543** during movement. These positions are only introduced to serve as a reference for movement, to clearly illustrate the positions of the pressing member **541**, the locking portion **34**, the clamped member **543** relative to the first structural body **10** during the process of mounting the second structural body **30** to the first structural body **10**, and these positions may not be uniquely determined. For example, in the case of engineering errors, these positions should be within a suitable area on the first structural body **10**, and should not be construed as strictly limiting the technical solutions.

(81) For example, when the second structural body **30** is not mounted to the electronic device **500**, or the second structural body **30** has already been mounted to the electronic device **500** and may be normally used, it may be considered that the pressing member **541** is located at the first position. Normally, the pressing member **541** cannot move towards the outside of the electronic device **500** along the first direction. For another example, the first structural body **10** may be provided with a position-limiting structure, and in the second position, the pressing member **541** abuts against the position-limiting structure. Normally, the pressing member **541** cannot continue moving towards the inside of the electronic device **500** along the first direction, thereby preventing the pressing member **541** from being excessively pressed and damaged. The first structural body **10** may be provided with corresponding position-limiting structures, so that the pressing member **541** abuts against such position-limiting structures in the first position and the second position respectively, and thus the first position and the second position mentioned above may be determined.

(82) Certainly, the first position and the second position may be determined in other ways. For example, in the above embodiments, when the pressing member **541** is in the first position, the end surface of the pressing member **541** exposed from the housing **14** is flush with the end surface of

the housing **14** at the key groove **1011**, and such appearance characteristic may be used to determine the first position of the pressing member **541**. For another example, when the pressing member **541** is pressed to enable the second structural body **30** to be detached from the first structural body **10**, it may be considered that the pressing member **541** is located at the second position.

(83) Similarly, the third position and the fourth position may also be determined in multiple ways. For example, when the second structural body **30** is not mounted to the electronic device **500**, or the second structural body **30** has already been mounted to the electronic device **500** and may be normally used, it may be considered that the clamped member **543** is located at the third position. Normally, the pressing member **541** cannot move along the first direction to get closer to the pressing member **541**. In the process of mounting the second structural body **30** to the first structural body **10**, another extreme position of the movement of the clamped member **543** along the first direction is the fourth position. If the clamped member **543** is fixed at the fourth position relative to the first structural body **10**, the locking portion **34** of the second structural body **30** can smoothly get in and out the groove **1031**. For another example, in the aforementioned embodiment, the third position and the fourth position may also be determined through the mating of the protrusion **5439** of the clamped member **543** and the holding hole **127**.

(84) In the locking structure provided by the embodiments of the present disclosure, the key is configured to be connected between the first structural body and the second structural body. While the key is provided at the first structural body, the clamped member of the key can be engaged with the locking portion of the second structural body. By making the clamped member movably engaged with the locking portion, the second structural body can be conveniently connected to and detached from the first structural body. For example, in the process of mounting the second structural body to the first structural body, the clamped member is driven by the locking portion to move relative to the first structural body, and is restored so as to be engaged with the locking portion; as such, the second structural body is enabled to be conveniently mounted to the first structural body. For another example, in the process of detaching the second structural body from the first structural body, the pressing member can be pressed to drive the clamped member to move so as to be disengaged from the locking portion; as such, the second structural body is enabled to be conveniently detached from the first structural body. When the locking structure is applied to a smart wearable device, a detachable connection of the device body with a wearable part can be conveniently achieved by means of the key structure.

(85) Further, in the above locking structure, the pressing member and the clamped member of the key are positioned at the first structural body through the first mounting portion and the second mounting portion, respectively. The pressing member and at least partial structure of the clamped member are arranged opposite to each other, and the pressing member can be pressed and drive the clamped member to move. Thus, transmission of movement is enabled between the pressing member and the clamped member by means of the opposite position relationship thereof, and the positioning and mounting of the pressing member can be are performed separately from the positioning and mounting of the clamped member. This avoids a fitting issue in installation that is caused due to a too long dimension chain generated when the pressing member and the clamped member are fixedly connected through other fastening structures. In this way, the key is designed as a pressing member and a clamped member which are substantially separated, which breaks the original dimension chain of the key. In addition, there is no longer a dimension chain for assembly between the pressing member and the clamped member, and the pressing member and the clamped member may be respectively provided at the first mounting portion and the second mounting portion of the first structural body. There is thus a short dimension chain generated when the key is mounted onto the first structural body, which reduces the difficulty of producing and assembling the locking structure. When the locking structure is applied to a smart wearable device or/and a smart watch, it is beneficial to improve the accuracy installation of the key, and ensure the yield of

installation of the parts as well as the appearance consistency for the smart wearable device or the smart watch.

(86) Based on the foregoing locking structure, an embodiment of the present disclosure further provides a smart wearable device. The smart wearable device provided in the embodiment of the present disclosure is a portable device that is directly worn on a person or integrated into clothing or accessories of a user. The smart wearable device may include, but is not limited to, a watch, a smart bracelet, a smart wristband, smart glasses, a finger ring, a helmet, and the like. For ease of explanation, it is illustrated in detail below by taking a smart watch as an example. The watch strap provided by the embodiments of the present disclosure may be adaptively adjusted according to different positions where the wearable device is worn.

(87) As illustrated in FIG. **17**, an embodiment of the present disclosure provides an electronic device **500** having a locking structure **100**. The electronic device **500** includes a first structural body **10** and electronic components, such as a circuit board (not shown) and a battery (not shown), provided on the first structural body **10**. The first structural body **10** may include a middle frame **12** and a housing **14** connected to the middle frame **12**. The middle frame **12** and the housing **14** together enclose a receiving space **131**. The receiving space **131** is configured to receive the electronic components such as the circuit board and the battery of the electronic device **500**. After the electronic device **500** is worn on the wrist of the user, at least part of the surface of the housing **14** is attached to the wrist of the user. The middle frame **12** may be made of a non-metal material such as plastic, rubber, silicon, wood, ceramic or glass, and the middle frame **12** may also be made of a metal material such as stainless steel, aluminum alloy or magnesium alloy. The middle frame **12** may also be a metal injection molded part, that is, a metal material is used to ensure the structural rigidity of the middle frame **12**, and a mounting and positioning structure, such as a protrusion, a groove or a threaded hole, may be provided on the inner surface of the metal body through injection molding. The housing **14** may be made from a glass or ceramic or plastic material, and the material of the housing **14** may be the same as or different from that of the middle frame **12**. In some embodiments, the middle frame **12** may be integrally formed with the housing **14**.

(88) As illustrated in FIG. **18**, an embodiment of the present disclosure provides a smart wearable device **300** having a locking structure **100**. The smart wearable device **300** may be, but is not limited to, a watch, a smart bracelet, a smart wristband, smart glasses, a finger ring, a helmet or other smart wearable devices. The smart wearable device **300** of the present embodiment is described by taking a smart watch as an example, and the locking structure **100** is described by taking a connection structure between the watch strap and the watch body of the smart watch as an example. The smart wearable device **300** includes a device body **200**, a locking structure **100**, and a wearable part. The wearable part **400** is connected to two sides of the device body **200** through the locking structure **100** for the user to wear.

(89) In this embodiment, the device body **200** is a watch body, and the watch body may be a main body capable of implementing a function of a smart watch. The first structural body **10** of the lock structure **100** is provided at the device body **200**. The device body **200** may include a display screen **201**. The display screen **201** is connected to the housing **14**, and a cavity is defined jointly by the display screen **201** and the housing **14**. Components such as a chip, a sensor, and a battery may be provided inside the cavity.

(90) The sensor includes, but is not limited to, a temperature sensor capable of detecting a body temperature, a vibration sensing sensor or a photoelectric sensor capable of detecting heart rate, or a pressure sensor for detecting blood pressure, and the like, so that the sensor can detect various health indicators of the wearer's body, such as the heart rate, body temperature or blood pressure. The sensor may also include an image sensor, a visible light sensor, an infrared light sensor, or the like. The chip may include one or more processing units, for example: an Application processor (AP), a modem processor, a memory, a Digital signal processor (DSP), a baseband processor,

and/or a Neural-network processing unit (NPU). The display screen **201** may be a Liquid Crystal Display (LCD) or an Organic Light-Emitting Diode (OLED) display. The display screen **201** may be configured to display various information such as time, health indicators, and information. Certainly, the display screen **201** may be a touch screen, or an operating component such as a key may be provided on the display screen **201**. The battery may supply power to components such as the display screen **201**, the controller, and the processor. The battery includes, but is not limited to, a lithium battery, a dry battery, a storage battery, and the like.

(91) In the embodiment of the present disclosure, the device body **200** may further include a wireless communications module. The wireless communications module may provide a solution for wireless communications applied to the smart watch, including a Wireless local area networks (WLAN), a Bluetooth (T), Frequency modulation (FM), an Infrared (IR) technique, or the like. The wireless communication module may implement a communication connection between the smart wearable device **300** and an external device, such as a mobile phone, or a computer.

(92) The wearable part **400** is a structure for the user to wear. The second structural body **30** of the locking structure **100** is provided on the wearable part **400**. In the embodiment, the wearable part **400** includes a first wearable part **410**, a second wearable part **430** and a locking portion **450**. The first wearable part **410** and the second wearable part **430** are respectively connected to two opposite sides of the device body **200** through the locking structure **100**. The locking portion **450** is used to connect the first wearable part **410** and the second wearable part **430** together. As such, the first wearable part **410**, the second wearable part **430** and the device body **200** together define a substantially annular structure for the user to wear. It is notable that, in the specification of the present disclosure, when one component is considered to be "provided at" another component, it may be connected to or directly provided at another component, or there may be a middle component (that is, the two components are indirectly connected). When one component is considered to be "connected to" another component, it may be directly connected to another component, or there may be a middle component, that is, the two components may be indirectly connected.

(93) In the embodiments of the present disclosure, there are two wearable parts **400** (one of which is shown). Two opposite ends of the device body **200** each are provided with the groove **1031**. Each of the two wearable parts **400** has one end thereof connected with the device body **200**, and another end of each the two wearable parts **400**, which is away from the device body **200**, may be buckled to define a receiving space, so that the device body **200** may be worn to the wrist of the user through the wearable parts **400**. In other embodiments, the wearable part **400** may be in a one-piece structure. One end of the wearable part **400** is connected to one end of the device body **200**, the other end of the device body **200** may be provided with a retaining ring for the wearable part **400** to pass through, and the free end of the wearable part **400** may pass through and wound around the retaining ring, and then be fixed to another position of the wearable part **400** to define a receiving space. The size of the receiving space is easy to adjust, so as to facilitate the user to wear.

(94) In the smart wearable device and the locking structure thereof provided by the embodiments of the present disclosure, the key is configured to be connected between the first structural body and the second structural body. While the key is provided at the first structural body, the clamped member of the key can be engaged with the locking portion of the second structural body. By making the clamped member movably engaged with the locking portion, the second structural body can be conveniently connected to and detached from the first structural body. For example, in the process of mounting the second structural body to the first structural body, the clamped member is driven by the locking portion to move relative to the first structural body, and is restored so as to be engaged with the locking portion; as such, the second structural body is enabled to be conveniently mounted to the first structural body. For another example, in the process of detaching the second structural body from the first structural body, the pressing member can be pressed to drive the clamped member to move so as to be disengaged from the locking portion; as such, the second

structural body is enabled to be conveniently detached from the first structural body. When the locking structure is applied to the smart wearable device, a detachable connection of the device body with a wearable part can be conveniently achieved by means of the key structure.

(95) Based on the above locking structure **100** and the smart wearable device **300**, an embodiment of the present disclosure further provides a housing assembly (not shown in the figure). The housing assembly includes the housing **14**, the middle frame **12**, and the key **54** mentioned above. The housing **14** is provided with the first mounting portion **101**, and the middle frame **12** is provided with the second mounting portion **103**. The key **54** is connected to the middle frame **12**, and is configured to connect the middle frame **12** and an external structure. The key **54** includes the pressing member **541** and the clamped member **543**. The pressing member **541** is movably provided at the first mounting portion **101**, and is configured to move relative to the first mounting portion **101** along a first direction under an external force. The clamped member **543** is movably provided at the second mounting portion **103** and is positioned at the middle frame **12** through the second mounting portion **103**, and the clamped member **543** is configured to be engaged with an external structure (such as the locking portion **34**). In the process of mounting the external structure to the middle frame **12**, the external structure moves along the second direction to push the clamped member **543** to move relative to the middle frame along the first direction, and the clamped member can be restored so as to be engaged with the external structure. In the process of detaching the external structure from the middle frame **12**, the pressing member **541** can be pressed to drive the clamped member **543** to move along the first direction so as to be disengaged from the external structure, where the first direction is different from the second direction. In this way, the middle frame **12**, the housing **14**, the pressing member **541** and the clamped member **543** can form a modular assembly, and when being applied, the modular assembly can be directly connected with other components of an actual application product, and the locking structure **100** can thus be assembled, which improves the assembly efficiency of the assembly. In other embodiments, the housing assembly may further include the features and structures introduced in any one of the above embodiments, and details thereof are not repeated here.

(96) Based on the locking structure **100** and the smart wearable device **300** mentioned above, an embodiment of the disclosure further provides an electronic device. The electronic device may be, but is not limited to, a smart device such as a watch, a smart bracelet, a smart wristband, smart glasses, a finger ring, a helmet, or the like, or a smart portable communication device. The electronic device may be installed with a strap, and the strap may has any one of features of the second structural body provided in the above embodiments or a combination thereof, which will not be repeated here. The electronic device comprises a casing, a key and a clamped member. The casing may have any one of features of the first structural body provided in the foregoing embodiments or a combination thereof, and details thereof are not repeated here. The casing is provided with a first mounting portion and a second mounting portion communicated with each other. The key includes a pressing member and an elastic connection member. The elastic connection member is connected to the pressing member, and the pressing member is provided at the first mounting portion. The clamped member is provided at the second mounting portion. The assembling and positioning of the clamped member to the casing are performed separately from the assembling and positioning of the pressing member to the casing. The clamped member is used to mate with the strap, to make the strap confined at the casing. When the strap is confined at the casing, the elastic connection member abuts against the casing, so that the pressing member is confined at the casing. When the pressing member is pressed, the pressing member moves relative to the casing, and the pressing member drives the clamped member to move, so that the clamped member is disengaged from the strap, and the strap can be detached from the casing.

(97) In this specification, specific features or characteristics described may be combined in any one or more embodiments or examples in a suitable manner. Furthermore, various embodiments or examples described in this specification, as well as features of different embodiments or examples,

may be combined by those skilled in the art without contradiction. Finally, it is notable that the above embodiments are only used to describe the technical solutions of the present disclosure, rather than limiting such technical solutions. Although the present disclosure has been described in detail with reference to the foregoing embodiments, those skilled in the art will understand that they still can modify the technical solutions described in the foregoing embodiments, or equivalently substitute some of the technical features; however, such modifications or substitutions do not drive the essence of the corresponding technical solutions to depart from the spirit and scope of the technical solutions of the embodiments of present disclosure.

Claims

1. A locking structure, comprising: a first structural body, provided with a first mounting portion and a second mounting portion; a second structural body, comprising a main body and a locking portion protruding relative to the main body; a pressing member, wherein the pressing member is movably provided at the first mounting portion, and is configured to move relative to the first mounting portion under an external force; and a clamped member, wherein the clamped member is movably provided at the second mounting portion, and is positioned at the first structural body through the second mounting portion; the clamped member is configured to be engaged with the locking portion; and at least partial structure of the clamped member is arranged opposite to the pressing member; wherein in a process of mounting the second structural body to the first structural body, the clamped member is adapted to be driven by the locking portion to move relative to the second mounting portion, and be restored so as to be engaged with the locking portion; and in a process of detaching the second structural body from the first structural body, the pressing member is capable of being pressed to drive the clamped member to move, so as to make the clamped member disengaged from the locking portion.
2. The locking structure of claim 1, wherein one end of the pressing member is overlapped with the at least partial structure of the clamped member; and wherein the one end of the pressing member is arranged opposite to and spaced apart from the at least partial structure of the clamped member.
3. The locking structure of claim 2, wherein the locking structure further comprises a buffer component; the buffer component is provided between the pressing member and the clamped member, and is configured to push the clamped member to move when the pressing member is pressed; wherein one side of the buffer component is connected to one of the pressing member and the clamped member, and another side of the buffer component is spaced apart from the other one of the pressing member and the clamped member; or wherein one side of the buffer component is connected to one of the pressing member and the clamped member, and the other side of the buffer component is connected to the other one of the pressing member and the clamped member.
4. The locking structure of claim 3, wherein the buffer component is provided with a first connection portion, the clamped member is provided with a second connection portion, and the first connection portion and the second connection portion are adapted to mate with each other so as to fix the buffer component relative to the clamped member; one of the first connection portion and the second connection portion is a groove, and the other one of the first connection portion and the second connection portion is a protrusion capable of mating with the groove.
5. The locking structure of claim 1, wherein the first structural body is further provided with a position-limiting portion; the locking structure further comprises an elastic connection member connected to the pressing member, the elastic connection member is provided with a positioning portion, and the positioning portion is configured to mate with the position-limiting portion; and in a process of mounting the pressing member to the first mounting portion, the elastic connection member is capable of being squeezed and deformed by the first structural body, until the positioning portion and the position-limiting portion mate with each other to define a position of the pressing member relative to the first structural body.

6. The locking structure of claim 5, wherein the first mounting portion comprises an accommodating groove provided in the first structural body, the pressing member is movably accommodated in the accommodating groove, and the positioning portion and the position-limiting portion are adapted to movably mate with each other to enable the pressing member to move in the accommodating groove in a first direction; and in the process of mounting the second structural body to the first structural body, the locking portion is adapted to move in a second direction to drive the clamped member to move relative to the first structural body in the first direction; the first direction is perpendicular to the second direction.
7. The locking structure of claim 6, wherein the position-limiting portion is a groove, the groove has a bottom wall, and the positioning portion is accommodated in the groove and arranged opposite to the bottom wall; the first structural body is provided with a projection located in the accommodating groove, the projection is arranged opposite to at least part of the pressing member; and the bottom wall and the projection are configured to limit movement of the pressing member in the accommodating groove.
8. The locking structure of claim 5, wherein the elastic connection member further comprises a fixing portion, and the fixing portion is connected to the pressing member; the positioning portion is movably connected to the fixing portion, and is capable of protruding relative to the pressing member; in the process of mounting the pressing member to the first mounting portion, the positioning portion is capable of being squeezed by the first structural body to retract relative to the fixing portion, and is capable of extending out relative to the fixing portion to mate with the position-limiting portion, to define the position of the pressing member relative to the first structural body; and wherein the fixing portion is provided with an accommodating cavity, the positioning portion comprise two positioning portions, and the two positioning portions are respectively arranged at opposite ends of the accommodating cavity, and each of the two positioning portions is capable of extending out of the fixing portion through the accommodating cavity, to protrude relative to a perspective one of two ends of the pressing member; the elastic connection member further comprises an elastic portion, and the elastic portion is provided between the two positioning portions.
9. The locking structure of claim 1, wherein the locking structure further comprises a position-restoring member, and the position-restoring member is provided between the clamped member and the first structural body; and in the process of detaching the second structural body from the first structural body, the pressing member is capable of being pressed to drive the clamped member to move, and the position-restoring member is compressed to apply, to the clamped member, a supporting force towards the pressing member.
10. The locking structure of claim 9, wherein the pressing member comprises a pressing portion and a projecting portion; the pressing portion is provided at the first mounting portion; the projecting portion is connected to a side of the pressing portion facing towards the clamped member, and projects out relative to the pressing portion; the clamped member comprises an abutting portion and a clamped portion connected to the abutting portion; the abutting portion is spaced apart from or abuts against the projecting portion, and the clamped portion is spaced apart from the pressing portion; and when the second structural body is mounted to the first structural body, the locking portion is located between the clamped portion and the pressing portion and is engaged with the clamped portion; and wherein the locking portion is provided with a locking groove; in the process of mounting the second structural body to the first structural body, the clamped portion is adapted to be driven by the locking portion to move relative to the first structural body, and be restored so as to be at least partially accommodated in the locking groove, and be engaged with the locking portion; and in the process of detaching the second structural body from the first structural body, the pressing member is capable of being pressed to drive the clamped portion to be disengaged from the locking groove.
11. The locking structure of claim 9, wherein the position-restoring member comprises a first

magnet and a second magnet, the first magnet is connected to the clamped member, and the second magnet is connected to the first structural body; and in a process of mounting the locking portion to the first structural body, the first magnet and the second magnet are adapted to approach each other and generate a magnetic repulsion force.

12. The locking structure of claim 10, wherein a side of the locking portion facing away from the main body is provided with a first guiding slope; and a side of the clamped portion is provided with a second guiding slope; and wherein in the process of mounting the second structural body to the first structural body, the locking portion is adapted to push the second guiding slope of the clamped portion to drive the clamped member to move relative to the second mounting portion.

13. The locking structure of claim 1, wherein the first structural body comprises a middle frame and a housing, the housing is connected to the middle frame, the first mounting portion is provided at the housing, and the second mounting portion is provided at the middle frame; wherein the housing comprises an outer surface and an inner surface facing away from each other, and a side wall surface connected between the outer surface and the inner surface; the first mounting portion comprises an accommodating groove provided in the housing, and the accommodating groove runs through the inner surface and the outer surface, and extends to the side wall surface; and wherein the pressing member is movably provided in the accommodating groove, and a surface of the pressing member is coplanar with the outer surface of the housing.

14. The locking structure of claim 13, wherein the second mounting portion comprises a receiving space provided in the middle frame; the middle frame comprises a bearing surface and a side surface connected with the bearing surface; the receiving space runs through the bearing surface to define a first opening, and runs through the side surface to define a second opening; the housing is covered on the bearing surface; the middle frame further comprises a boundary portion, the boundary portion is covered on the side surface and partially covered on the second opening; and the locking portion is adapted to pass through the second opening so as to be engaged with the clamped member.

15. The locking structure of claim 13, wherein the middle frame is provided with a first positioning portion, the clamped member is provided with a second positioning portion, the first positioning portion and the second positioning portion are adapted to mate with each other, to enable the clamped member to be positioned at the middle frame and have a movement degree of freedom in a predetermined direction; and wherein one of the first positioning portion and the second positioning portion comprises a guiding structure, the guiding structure is arranged along the predetermined direction; the other one of the first positioning portion and the second positioning portion comprises a sliding component, and the sliding component is slidably connected to the guiding structure.

16. The locking structure of claim 15, wherein a side of an abutting portion of the clamped member facing away from the second positioning portion is provided with a receiving groove; at an end of the abutting portion facing away from the pressing member, the receiving groove extends to an end surface of the abutting portion; and an end of a position-restoring member is accommodated in the receiving groove.

17. The locking structure of claim 13, wherein the second mounting portion comprises a holding hole provided in the middle frame, the clamped member is provided with a holding protrusion, the holding protrusion is movably accommodated in the holding hole, and a size of the holding protrusion in a predetermined direction is smaller than a size of the holding hole in the predetermined direction.

18. The locking structure of claim 1, wherein the locking structure further comprises a baffle detachably connected to the first structural body, the baffle is provided at the second mounting portion and adapted to shield at least part of the clamped member, and the baffle is provided with a through hole for mounting the locking portion.

19. A housing assembly, comprising: a housing provided with a first mounting portion; a middle

frame provided with a second mounting portion; and a key connected to the middle frame, wherein the key is configured to connect the middle frame and an external structure comprising a locking portion, and the key comprises: a pressing member, wherein the pressing member is movably provided at the first mounting portion, and is configured to move relative to the first mounting portion in a first direction under an external force; and a clamped member, wherein the clamped member is movably provided at the second mounting portion, and positioned at the middle frame through the second mounting portion, and the clamped member is configured to be engaged with the locking portion of the external structure; at least partial structure of the clamped member is arranged opposite to the pressing member; wherein in a process of mounting the external structure to the middle frame, the external structure is adapted to move in a second direction to push the clamped member to move relative to the middle frame in the first direction, and the clamped member is capable of being restored so as to be engaged with the locking portion of the external structure; in a process of detaching the external structure from the middle frame, the pressing member is capable of being pressed to drive the clamped member to move in the first direction so as to be disengaged from the external structure, wherein the first direction is different from the second direction.

20. An electronic device configured to be installed with a strap, the electronic device comprising: a casing, provided with a first mounting portion and a second mounting portion communicated with each other; a key comprising a pressing member and an elastic connection member, wherein the elastic connection member is connected to the pressing member, and the pressing member is provided at the first mounting portion; and a clamped member provided at the second mounting portion, wherein assembling and positioning of the clamped member to the casing are performed separately from assembling and positioning of the pressing member to the casing; wherein the clamped member is adapted to mate with a locking portion of the strap, to make the strap confined at the casing; when the strap is confined at the casing, the elastic connection member abuts against the casing, to make the pressing member confined at the casing; the pressing member is adapted to, when being pressed, move relative to the casing and drive the clamped member to move, to enable the clamped member to be disengaged from the strap, and enable the strap to be detached from the casing.